

Question Paper

Exam Date & Time: 27-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Cloud Computing [NPT_4120]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyse how the NIST definition of cloud computing reflects its five essential characteristics. (5)
Explain how these characteristics collectively differentiate cloud computing from traditional IT infrastructure.
- A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Differentiate between public, private, hybrid, and community cloud deployment models by analysing their suitability for different types of organizations. (3)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- C) Illustrate with suitable examples how IaaS, PaaS, and SaaS are used in practical cloud computing environments. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Evaluate the importance of measurable SLA metrics in maintaining accountability between cloud service providers and customers. (5)

- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Assess how different pricing models influence the budgeting and scalability strategies of cloud-based organizations. (3)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- C) Demonstrate how automation tools can help overcome performance and audit challenges in multi-tenant cloud environments. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) Evaluate the role of automation in maintaining accuracy and compliance across all stages of the cloud data lifecycle. (5)

- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Explain how organizations use data governance policies to meet industry-specific compliance standards in cloud computing. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Describe how object storage simplifies data retrieval and scalability in large-scale cloud environments. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Discuss how predictive analytics can be applied for proactive resource allocation in cloud environments. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Evaluate how adaptive load balancing strategies maintain optimal performance during traffic spikes in cloud systems. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Explain how energy-efficient virtualization techniques contribute to sustainable cloud data center operations. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Investigate the impact of shared responsibility models on addressing security challenges in cloud environments. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Apply encryption and key management techniques to secure data transmission between cloud clients and servers. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Evaluate the role of IAM in enforcing least-privilege access and preventing insider threats in cloud systems. (2)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

NPT_4120:Cloud Computing

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

NPT_4120:Cloud Computing

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	9(18%)

4	Analysis	11	41(82%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)